

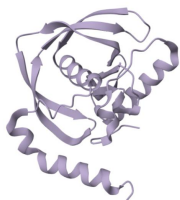
PFArena: How to obtain better mutants?

T1

Single mutant generation

Aim: Good single-mutant

INPUT :



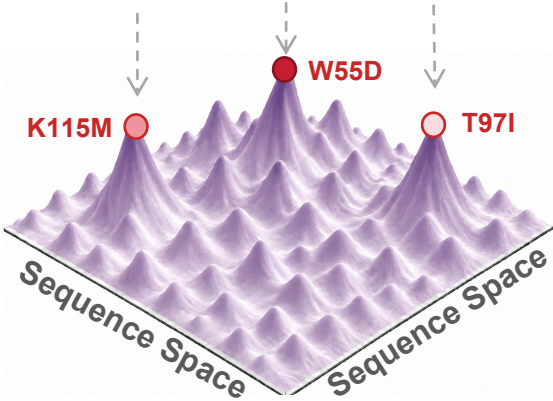
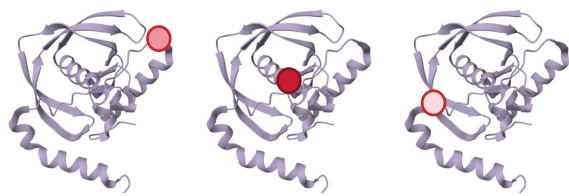
WT



Measurement-free

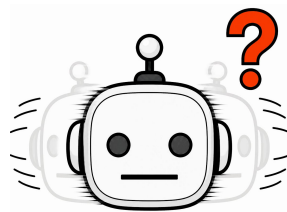
OUTPUT :

Best candidates



Which multi-mutant is the best choice?

Which single mutation should I test first?



PLM / LLM / Agent



What improves my measured anchor?

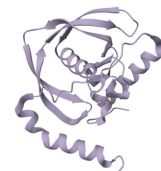
How should I use my DMS map?

T2

Measurement-free multi-mutant ranking

Aim: Good multi-mutant without supplementary info

INPUT :



WT

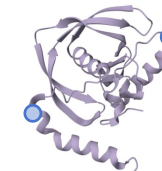


Candidate library



Measurement-free

OUTPUT :



1. V31H+K115M
2. V10I+D25G+A119T
3. E26K+D162N
4. R33S+G145D ...

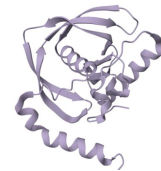
Ranked library

T3

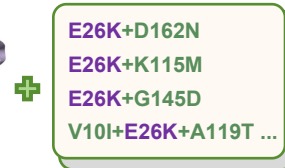
Anchor-informed multi-mutant ranking

Aim: Good multi-mutant based on one measured mutant (anchor)

INPUT :



WT



Candidate library



Anchor info

OUTPUT :



1. E26K+G145D
2. V10I+E26K+A119T
3. E26K+D162N
4. E26K+K115M ...

Ranked library

T4

Single-mutant-informed multi-mutant ranking

Aim: Good multi-mutant with sufficient single mutant info

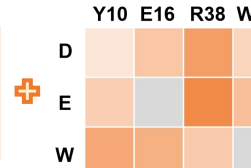
INPUT :



WT

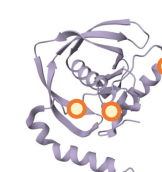


Candidate library



Mutation info

OUTPUT :



1. E16D+R38E+W55D
2. R38D+W55D
3. Y10W+R38D
4. R38E+W55D ...

Ranked library